

Lishui Fan

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Education

Zhejiang University, Computer Science and Technology

- College of Computer Science and Technology.
- Research focus: LLM-based code generation and software engineering.

Hangzhou, China

Jan 2023 – present

Experience

Baidu ERNIE Bot Team, Research Intern

Working on large language model applications and code intelligence for ERNIE Bot.

China

present

ByteDance Douyin AI4SE, Research Intern

Worked on AI for software engineering and code intelligence problems in practical development settings.

China

Taotian Future Life Lab, Research Intern

Conducted applied AI research for future-life and intelligent systems.

China

Research

LLM-based Code Generation

Studies training, evaluation, and optimization methods for code LLMs, including fault-guided fine-tuning, keyword-enhanced prompting, and reasoning-process reward learning.

- Code LLM capability evaluation.
- Code-change understanding.
- Repository-aware and agent-oriented software engineering.

Industry Collaboration

Core contributor to projects with Huawei and Tencent, translating practical software engineering needs into model training, evaluation, and optimization problems.

- CCF-Huawei Populus Grove Fund project.
- Tencent Rhino-Bird special project.
- Huawei industry collaboration project.

Publications

ReCode: Reinforcing Code Generation with Reasoning-Process Rewards

A CCF-A paper on reasoning-process reward learning for code generation.

Lishui Fan, Yu Zhang, Mouxiang Chen, Zhongxin Liu

arxiv.org/abs/2508.05170

ZeroCoder: Can LLMs Improve Code Generation Without Ground-Truth Supervision?

A preprint on label-free co-evolution of code and test generation.

Lishui Fan, Mouxiang Chen, Tingwei Zhu, Kui Liu, Xin Xia, Shanping Li, Zhongxin Liu

arxiv.org/abs/2604.07864

Balancing Latency and Accuracy of Code Completion via Local-Cloud Model Cascading

A collaborative Tencent-related paper on local-cloud model cascading for code completion.

Hanzhen Lu, Lishui Fan, Jiachi Chen, Qiuyuan Chen, Zhao Wei, Zhongxin Liu

arxiv.org/abs/2603.05974

Exploring the Capabilities of LLMs for Code-Change-Related Tasks

A CCF-A journal article studying LLMs for code-change-related tasks.

Lishui Fan, Jiakun Liu, Zhongxin Liu, David Lo, Xin Xia, Shanping Li

doi.org/10.1145/3709358

FGit: Fault-Guided Fine-Tuning for Code Generation

A CCF-A conference paper on fault-guided fine-tuning for code generation.

Lishui Fan, Zhongxin Liu, Haoye Wang, Lingfeng Bao, Xin Xia, Shanping Li

doi.org/10.1109/ASE63991.2025.00114

SEK: Self-Explained Keywords Empower Large Language Models for Code Generation

A CCF-A paper on keyword-enhanced code generation with LLMs.

Lishui Fan, Mouxian Chen, Zhongxin Liu

aclanthology.org/2025.findings-acl.324

Projects

CCF-Huawei Populus Grove Fund Project

Core research contributor to an industry-academic project on software engineering intelligence.

- Code LLM training and evaluation.
- Translation of real software engineering needs into research problems.

Tencent Rhino-Bird Special Project

Core contributor to a Tencent collaboration on code intelligence and code completion.

- Related collaborative work accepted to FSE 2026.
- Focus on latency-accuracy trade-offs for code completion.

Huawei Industry Collaboration Project

Core research contributor to a Huawei industry collaboration project.

- Practical software engineering scenarios.
- Code LLM optimization and evaluation.

Skills

Research Areas